MODULE 26 · SECTION 3 - STORING THE DATA

Knowledge Catalog and Federated Governance

The renames, the metadata model, glossary, discovery, profiling and quality, lineage, data products, lakes and zones, and data mesh with federated governance.

SECTION 3 of 5 EXAM WEIGHT 20% MODULE 26 of 45

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Find it, trace it, trust it. Then let each team own it, under shared rules.

One product, several names

- Dataplex became Dataplex Universal Catalog, now **Knowledge Catalog**, since 10 April 2026.
- The API, the CLI and IAM names are **unchanged**. It is still gcloud dataplex.
- **Data Catalog** is shut down, in phases from 1 June 2026. Knowledge Catalog replaces it.
- Exam questions may say Dataplex or Data Catalog. Read both as Knowledge Catalog.

REMEMBER

Existing Dataplex lakes, scans and settings keep working. Nothing needed migrating.

One catalog, six jobs

CAPABILITY	THE QUESTION IT ANSWERS
Catalog and search	What data do we have, and where is it?
Aspects and glossary	What does it mean, and who owns it?
Discovery	Which new files can become tables?
Profiling and quality	What does it look like, and is it right?
Lineage	Where did it come from, and what depends on it?
Data products	What is packaged, promised and ready to use?

THE PATTERN

Technical metadata is collected automatically. Business meaning is what people must add.

Entries, aspect types and aspects

- Every table, bucket or model has an **entry**, created automatically for Google Cloud sources.
- An **aspect type** is a template. An **aspect** is that template filled in on an entry or a column.
- **Required** aspects come with the entry type. Optional ones are added by people.
- System aspects include **overview**, **contacts** and **schema**. Scores land as aspects too.

CURRENCY CHECK

Gemini in BigQuery does not read custom aspects. Put meaning in descriptions, the overview aspect or glossary terms.

Aspect or glossary term?

ASPECT

Structured facts

- Fields from a template
- Owner, domain, retention class
- Same shape on every entry
- Searchable and filterable

GLOSSARY TERM

Shared meaning

- A business definition in words
- Such as 'active account'
- Linked to columns in many tables
- Synonyms and related terms

THE TIEBREAKER

'Who owns this, and how long do we keep it?' is an aspect. 'What does active mean here?' is a glossary term.

Turning new files into tables

- A **discovery scan** groups files in a bucket into tables, with schemas and partitions.
- It publishes a **BigQuery dataset** named after the bucket.
- With a **connection ID**: Lakehouse tables. Without one: plain external tables.
- Not for **Iceberg** or **Delta Lake** tables. Images and PDFs become object tables.

> THE TRAP
>
> Set the label to user_managed when you own a schema, or the next scan may change it.

Profiling and quality, in the catalog

- **Profiling scans** measure nulls, distinct values, ranges and top values.
- Profiles **recommend quality rules**, so nobody starts from a blank page.
- **Quality scans** publish a scorecard on the entry, and log results for alerts.
- Sensitive Data Protection results can land as aspects too, as in module 04.

> REMEMBER
>
> Module 06 compared quality scans with Dataform assertions. Scans watch data at rest. Assertions guard a pipeline.

Where it came from, and what depends on it

- Turn on the **Data Lineage API** per project, and supported services report by themselves.
- BigQuery copy, load and query jobs, within **24 hours**. Also Composer, Dataflow, Vertex AI.
- **Column-level** lineage for top-level BigQuery columns. Custom tools send **OpenLineage**.
- Lineage is kept for **30 days**, and it is billed.

> THE TRAP
>
> Thirty days of lineage is not a seven-year audit trail. Keep long-term evidence somewhere else.

Impact analysis before a change

01

Oracle billing

Datastream captures every change.

02

raw.accounts_changes

Append-only history, from module 24.

03

curated.accounts_current

The latest-state view.

04

Marts and dashboards

Everything that would break.

FOR LARKSPUR

Before renaming a column, open its lineage graph. Every arrow out is a consumer to warn.

Packaging data for others

- A **data product** packages assets for reuse: discoverable, trusted and accessible.
- It bundles documentation, aspects, and **access groups** mapped to Google Groups.
- A **contract** sets promises, such as refresh schedule and quality standards.
- Consumers, including **AI agents**, find it in search and request access from the owner.

FOR LARKSPUR

Westgrid's regional demand data becomes a data product, with a five-minute freshness promise in its contract.

Lakes, zones and assets

ENTITY	WHAT IT IS	FOR LARKSPUR
Lake	A domain, in one region	Metering, in europe-west2
Raw zone	Data in any format, from outside	Head-end files, as delivered
Curated zone	Structured, required formats, Hive layout	Clean Parquet readings
Asset	A bucket or a BigQuery dataset	gs://lk-lake-prod

HONEST LIMIT

Newer docs lead with the catalog and data products, but exam questions still ask about lakes and zones.

Who owns what, in a federated model?

EACH DOMAIN OWNS

Data as a product

- Its pipelines and its tables
- Publishing data products
- Quality against the contract
- Approving access requests

THE CENTRAL TEAM SETS

Shared rules

- Aspect types and the glossary
- Policies, locations and tags
- Quality thresholds to meet
- The self-serve platform itself

THE PATTERN

Domains own the data. The centre owns the standards. Neither owns both.

Phrases that pick the answer

THE QUESTION SAYS	REACH FOR
One definition of a business term	Glossary term, linked to columns
Owner and retention on **every table**	A required aspect type
New files, **no hand-written schemas**	A discovery scan, with a connection
What breaks if this column changes?	Column-level lineage
Domains **own**, standards stay **shared**	Data mesh, federated governance
A question mentions **Data Catalog**	Read it as Knowledge Catalog

THE PATTERN

Find, trace, trust, own: match the question to one of the four.

> Three Larkspur teams calculate 'active customer' in three different ways, and analysts cannot tell which tables use which.
>
> The data office wants one agreed definition, linked to the relevant columns across datasets, and findable in search.

What should the team use?

A. A glossary term in Knowledge Catalog, linked to those columns

B. A shared spreadsheet of definitions, linked from each dataset description

C. A label called active_customer, added to every table that uses it

D. One view per dataset that computes active customers the agreed way

◆ PAUSE AND DECIDE

What should the team use?

A Glossary term

One definition, linked to columns, and found in search.

B Spreadsheet

Not linked to columns, and not part of catalog search.

C Labels

A key and a value, with no definition behind them.

D A view per dataset

Computes a number, but does not define or link the term.

THE SIGNAL

One agreed business definition, linked to columns: a glossary term.

> Finance plans to change the meaning of amount_gbp in curated.bills, and needs to know which tables and dashboards depend on that column.
>
> Pipelines are BigQuery SQL run by Composer, and the Data Lineage API is enabled.

What should the team do?

A Search INFORMATION_SCHEMA.JOBS by hand for queries that mention the column

B Ask every team to list what depends on the column in a shared sheet

C Turn on Data Access audit logs, and search them for the column name

D Open column-level lineage for the column in Knowledge Catalog

◆ PAUSE AND DECIDE

What should the team do?

A Search job history

Slow, partial, and easy to miss a query.

B Ask every team

Relies on memory, and goes out of date at once.

C Audit logs

Shows who read data, not what was built from it.

D Column-level lineage

Draws every downstream table that uses that column.

THE SIGNAL

What depends on this column: column-level lineage.

> Larkspur now has four domain teams: metering, billing, customer and grid. One central team builds every pipeline, and has become a bottleneck.
>
> Leadership wants domains to own their data, while keeping consistent standards for metadata, quality and access.

Which operating model fits best?

A Keep one central team, and hire more engineers into it

B Domains publish data products, under standards set centrally

C Give each domain its own organisation, with its own rules

D Copy every domain's data into one central project each night

◆ PAUSE AND DECIDE

Which operating model fits best?

A Grow the central team

Still one bottleneck, just a larger one.

B Federated governance

Domains own the data. The centre owns the standards.

C Separate organisations

Ownership without shared standards.

D Central copy

Duplicates data, and leaves ownership unclear.

THE SIGNAL

Domains own, standards stay shared: a data mesh with federated governance.

> Westgrid drops Parquet and CSV files into gs://lk-landing-prod every day, and new folders appear often.
>
> Analysts want each new folder queryable in BigQuery as a governed table, without copying files or writing schemas by hand.

What should the team set up?

A Write a CREATE EXTERNAL TABLE statement for each new folder

B A Cloud Run function that loads each new file into a native table

C A repeating discovery scan, configured with a connection

D A one-off Storage Transfer Service job into a BigQuery dataset

◆ PAUSE AND DECIDE

What should the team set up?

A Hand-written DDL

Exactly the schema writing the brief rules out.

B Load with a function

Copies every file into BigQuery storage.

C Discovery scan

Finds new folders, infers schemas, publishes Lakehouse tables.

D One-off transfer

Moves files once, and does not create tables.

THE SIGNAL

New files, no hand-written schemas, governed and in place: a discovery scan with a connection.

Module 26 in five lines

- Dataplex and Data Catalog questions now mean **Knowledge Catalog**.
- **Aspects** hold structured facts. **Glossary terms** hold shared meaning.
- **Discovery** turns files into tables. Profiles and quality scans build trust.
- **Lineage** shows impact, column by column, for 30 days.
- Data mesh: domains own **data products**, the centre owns the **standards**.

COMING UP

Next: module 27, the start of section four: serving BI, tools and precomputation.